Characterizing Challenging Behavior in Neurogenetic Syndromes: Exploring Patterns Across Methodologies

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Background

This study aims to examine challenging behavior profiles and families' idenitified treatment targets through two aims:

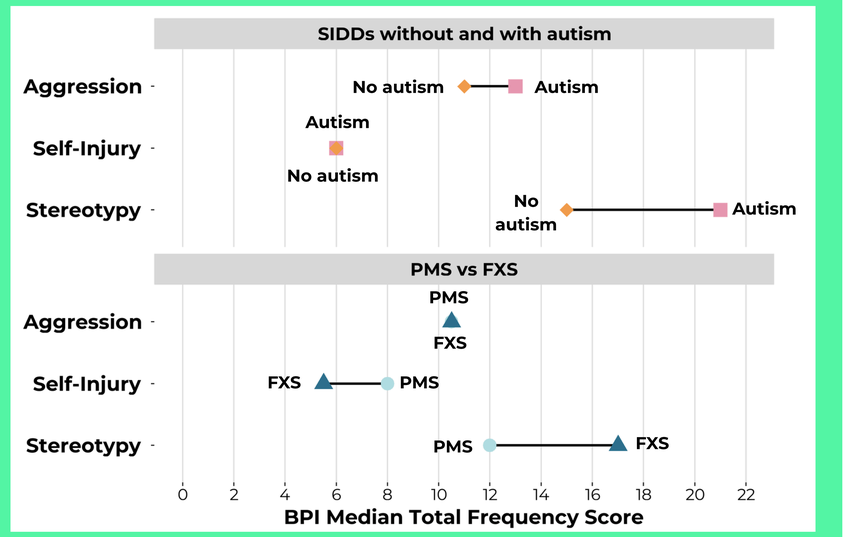
- Describing profiles and family priorities for treatment targets in children with **syndromic intellectual and developmental disabilities (SIDDs) with and without co-occurring autism spectrum disorder**.[1,2]
- Exploring syndrome-specific behavioral phenotypes which have been reported in the literature for children with **Phelan-McDermid syndrome (PMS)**[3] **and Fragile X syndrome (FXS)**.[4]

Note: Both person-first and identity-first language are used interchangeably throughout this poster to support clarity.

Procedure

- 52 caregivers (90.4% female) of eligible participants with SIDDs completed the **Behavior Problems Inventory-01 (BPI-01)**[5] during clinical screening and the **Parent Target Problems (PTP)**[6] with an Independent Evaluator (IE) during baseline according to established protocols (NCT06139172).
- **PTP** categorization:
  - Two independent reviewers categorized 106 **PTP** labels into 10 topography categories following methodology described in Sheridan et al., 2021.[7]
- Analyses:
  - **PTP** scores are presented as the percentage of families ranking the behavior target as priority 1, 2, or 3.
  - Median **BPI-01** subscale total frequency and severity scores across subgroups were calculated.
  - Mann-Whitney *U* tests were applied to evaluate group differences across **BPI-01** frequency and severity.

Figure 1. BPI Median Total Frequency Scores



Median *stereotypy* scores are significantly higher in children with **SIDDs with co-occurring autism.** No other significant group differences were found.

Table 1. Parent Target Problems Across Sample

Group	Parent Target Problem Topography Category, % (*n*)		
	1	2	3
All (*n* = 52)	Aggression, 53.8% (28/52)	Compliance Issues, 50% (26/52)	Compliance Issues-Tantrums, 17.3% (9/52)
SIDDs without autism (*n* = 27)	Aggression, 51.9% (14/27)	Compliance Issues, 48.1% (13/27)	Compliance Issues-Tantrums, 22.2% (6/27)
SIDDs with autism (*n* = 25)	Aggression, 56% (14/25)	Compliance Issues, 52% (13/25)	Safety Concerns, 24% (6/25)

Table 2. Parent Target Problems in PMS and FXS

Group	Parent Target Problem Topography Category, % (*n*)		
	1	2	3
PMS (*n* = 12)	Aggression, 58.3% (7/12)	Compliance Issues, 41.7% (5/12)	Safety Concerns, 33.3% (4/12)
FXS (*n* = 14)	Aggression, 42.9% (6/14)	Compliance Issues, 57.1% (8/14)	Hyperactivity/ Impulsivity, 28.6% (4/14)

The top two caregiver-reported targets were *Aggression* and *Compliance Issues* across all analyses on the **PTP**.

Results

Child Demographics (*n* = 52)

- **Age:** 2-12 years, *M* = 7.12, *SD* = 3.25
- **Sex assigned at birth:** Male: *n* = 38, 73.1%; Female: *n* = 14, 26.9%
- **Genetic Syndrome:** FXS: *n* = 14, 26.9%; PMS: *n* = 12, 23.1%; Angelman Syndrome (AS): *n* = 6, 11.5%; Other: *n* = 20, 38.5%
- **SIDDs with co-occuring autism:** *n* = 25, 48.1%
- **Intellectual Disability Level:** Borderline: *n* = 3, 5.8%; Mild: *n* = 25, 48.1%; Moderate: *n* = 17; 32.7%; Severe ID = *n* = 7, 13.5%

BPI-01 Subgroup Mann-Whitney *U* Analyses: Frequency and Severity Median Total Scores

- **SIDDs with autism compared to SIDDs without autism**
  - *Stereotypy* frequency (*U* = 227.5, *p* = .045) and severity (U = 193.0, *p* = .051) scores were marginally higher when autism was co-occurring.
  - No differences in frequency or severity were found in *Self-Injury* (Frequency: *U* = 322.0, *p* = .782; Severity: *U* = 191.5, *p* = .169) or *Aggression* (Frequency: *U* = 297.0, *p* = .463; Severity: *U* = 227.5, *p* = .305).
- **PMS and FXS**
  - No group differences were found in frequency or severity across all **BPI-01** domains: *Aggression* (Frequency: U = 84.0, p = 1.0; Severity: U = 66.0, p = .741), *Self-Injury* (Frequency: *U* = 78.0, *p* = .776; Severity: *U* = 28.0, *p* = .068), or *Stereotypy* (Frequency: *U* = 105.0, *p* = .291; Severity: *U* = 108.0, *p* = .108).

Discussion

- Caregiver priorities for intervention may not reflect challenging behaviors occurring at the highest severity or frequency.
  - *Aggression* and *Compliance* are most often selected as caregiver top concerns for intervention on the **PTP**
  - However, the highest median frequency and severity scores were found on the *Stereotypy* domain of the **BPI-01**
- Comparisons on the **BPI-01** suggest similarities in challenging profiles at the group level in **PMS** and **FXS**.
- **Co-occuring autism** is associated with elevated frequency and severity of *Stereotypy* compared to **SIDDs without autism**. However, as noted above, *Stereotypy* is less likely to be selected for intevention targets.

WINGS Website

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ClinicalTrials.gov NCT06139172

References